

ANTIBACTERIAL ACTIVITY OF IRON(II) AND ZINC(II)-AMINO ACID COMPLEXES

S. JAVAID KHURSHID

Biological Quality Control Section
Nuclear Chemistry Division
PINSTECH, P.O. Nilore, Islamabad

ABSTRACT

Zinc(II) and Iron(II) complexes of ten amino acids viz: L- alanine, L-arginine, glycine, L-histidine, L-phenylalanine, L-lysine, L-proline, L-threonine, L-tryptophan, L-tyrosine and L- leucine were prepared and their antibacterial activity was studied against the standard strains of *Escherichia coli* (ATCC 25922), *Staphylococcus aureus* (ATCC 25923) and *Streptococcus pyogenes* (ATCC 27853) in comparison with the action of pure amino acids and salts of Iron(II) and Zn(II). The complexes of Iron(II) and Zinc(II)-phenylalaninates, histidinates, tryptophanates and tyrosinates were found more active against these strains as compared Zinc acetate and Ferrous sulphate salts, while the respective pure amino acids showed no antibacterial activity. The Zinc(II) complexes of phenylalaninate, histidine, tryptophanate and tyrosinate were more active as compared to their complexes with Iron(II). A significant antibacterial activity of other complexes were also observed against all of the three strains.

INTRODUCTION

A variety of transition metal complexes of α -amino acids have been prepared and reported (Bender 1957, Hay 1967, Kroll 1954, Springer 1963). These transition metal complexes have shown clinical advantages over their simple inorganic salts. The exact mode of action of such complexes in human body is not clear but it is known that they provide required essential metalloelements and thus play a role strengthening the defensive mechanism of the body against certain disease conditions. These metalloelement- amino acid complexes act as a source of metalloelements in disease condition. Beside this it is a well known fact that such complexes has an edge over the pure salts as they are devoid of astringent taste and cause comparatively less gastric distress (Crosby 1992). The real mechanism of their action in such conditions is not clear. However, in the past some anti- inflammatory (AI) drugs have been in use which were basically developed as antibacterials for example gold complexes used in Rheumatoids

arthritis (RA), also inhibit growth of tuberculi bacilli. Copper has remained a metal of interest due to its significance in biological systems. Some Cu(II)-amino acid complexes have been found somewhat active against mycoplasma infections (Mukhammad 1985). Copper-amino acid complexes have shown to possess anti-inflammatory activity (Jackson 1978), while their antibacterial activity have also been earlier reported by us (Khurshid 1990).

Keeping in view the significance and effectiveness of copper- amino acid complexes in different conditions. We have tried to explore the potential of zinc(II) and Iron(II)-amino acid complexes against some selective strains. These complexes have been prepared by a well known method (Khurshid 1990 and Sorenson 1976) and tested for their antibacterial activity against *E. coli* (ATCC 25922), *Stap. aureus* (ATCC 25923) and *Strep.* (ATCC 27853) (Khurshid 1990).

EXPERIMENTAL

Materials

Methanol, Zinc acetate and Ferrous sulphate of E. Merck and Amino acids (L-alanine, L-arginine, glycine, L-histidine, L- phenylalanine, L-lysine, L-proline, L-threonine, L-tryptophan, L- tyrosine and L-leucine) of Sigma Chemicals, were used. The strains of *Escherichia coli* (ATCC 25922), *Staphylococcus aureus* (ATCC 25923), *Streptococcus pyogens* (ATCC 27853) were obtained from National Institute of Health (NIH), Islamabad.

Method

Zinc(II) and Fe(II)-amino acid complexes were prepared in solution form according to the usual methods (Abernalden 1927, Sorenson 1976) by mixing 2m mol of amino acid and 1m mol of metal salt in 95% alcohol. The mixture was concentrated to one forth of its original volume and chilled to 0°C to obtain crystals. These crystals were filtered through sintered glass crucible and dried under vacuum over P₂O₅. A solution of each complex (10 ug/10 ul) was prepared in water and used for testing antibacterial activity. In each case 10 ug/10 ul of the complex solution was loaded on a susceptibility disc, dried and the antibacterial activity was measured against *S. aureus*, *S. pyogens* and *E. coli* in Blood Agar and Muller Hinton media. Ampicillin (10 ug) disc was used as standard according to the standard susceptibility testing procedures (Bauer 1966). The sensitivity of two metal salts and of all the amino acids (10 ug/10 ul each), was also measured similarly.

RESULTS AND DISCUSSION

The sensitivity results obtained are given in Table 1. All the amino acid were

found inactive against all the three strains. The activity shown by Iron(II) and Zinc(II)-histidine, leucine, and phenylalanine was more than the other metal-amino acid complexes, while the Zinc(II)-complexes were more active compared to Iron(II)-complexes. The activity shown by most of the other Iron(II) and Zinc(II)-amino acid complexes was greater than that of their salts indicating their significant clinical superiority. These results support the hypothesis postulating the partial role of antibiotic activity of AI drugs in combating inflammation which may be substantive. The action of these complexes is thought to be in part due to *in-vivo* formation of metalloelement amino acid complexes (May 1977). The knowledge of antibacterial activity of Iron(II) and Zinc(II)-amino acid complexes gained through antibacterial testing of these complexes may help to understand defensive mechanism in the body against certain diseases conditions and will provide an insight into the mode of action of other transition metal-amino acid complexes which are found in quite abundance in plasma and are known as low-molecular-weight complexes (May 1977). The role of transition metals in connective tissue metabolism is well established so there may be some probable involvement in connective tissue diseases. The recent computer models have aided study upto 10 metals ions with about 50 endogenous low-molecular-weight ligands and thus more than 10,000 possible complex species can be calculated (May 1977). The biochemistry of how the immune system is affected and exact detail about the plasma trace metal rises and depletion to low-molecular-weight complexes in the biofluid are not clearly accounted for. Under these circumstances the greater one's knowledge of the detailed bio-inorganic chemistry involved, the safer the design and administration of a new drug will become, for e.g. gold used in treatment of rheumatoid arthritis inhibited the growth of *Tuberculli bacilli* similar to some standard RA drugs (May, 1977). Some of the copper(II)-complexes have also shown activity against mycoplasma infection. Our results beside showing good antibacterial activity, support medicinal importance of these metalloelement-amino acid complexes in certain disease conditions and possible effectiveness in inflammation i.e. rheumatoid arthritis, as beside copper zinc also helps in healing cellular injury.

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Table 1
Antibacterial activity of Iron(II) and Zinc(II)-amino acid complexes

Compounds	Susceptibility Zone (mm)		
	<i>Esh. Coli.</i>	<i>Steph. Aureus</i>	<i>Strep. Pyogens</i>
Ampicillin	17	26	22
Iron Sulphate	8	9	9
Zinc Sulphate	8	8	9
Alaninate	I	I	I
Argininate	I	I	I
Glycinate	I	I	I
Histidinate	I	I	I
Leucinate	I	I	I
Methioninate	I	I	I
Phenylalaninate	I	I	I
Theroninate	I	I	I
Tryptophanate	I	I	I
Tyrosinate	I	I	I
Iron(II)-Alaninate	10	12	11
Zinc(II)-Alaninate	11	12	1
Iron(II)-Argininate	10	10	10
Zinc(II)-Argininate	10	11	10
Iron(II)-glycinate	11	10	10
Zinc(II)-glycinate	11	12	12
Iron(II)-histidinate	13	13	13
Zinc(II)-histidinate	14	15	14
Iron(II)-leucinate	11	11	10
Zinc(II)-leucinate	11	11	11
Iron(II)-Methioninate	10	11	10
Zinc(II)-Methioninate	11	10	10
Iron(II)-Phenylalaninate	12	11	13
Zinc(II)-Phenylalaninate	14	15	14
Iron(II)-Theroninate	11	11	10
Zinc(II)-Theroninate	11	12	11
Iron(II)-Tryptophanate	12	13	12
Zinc(II)-Tryptophanate	14	14	13
Iron(II)-Tyrosinate	12	12	12
Zinc(II)-Tyrosinate	13	14	14

I = Inactive

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